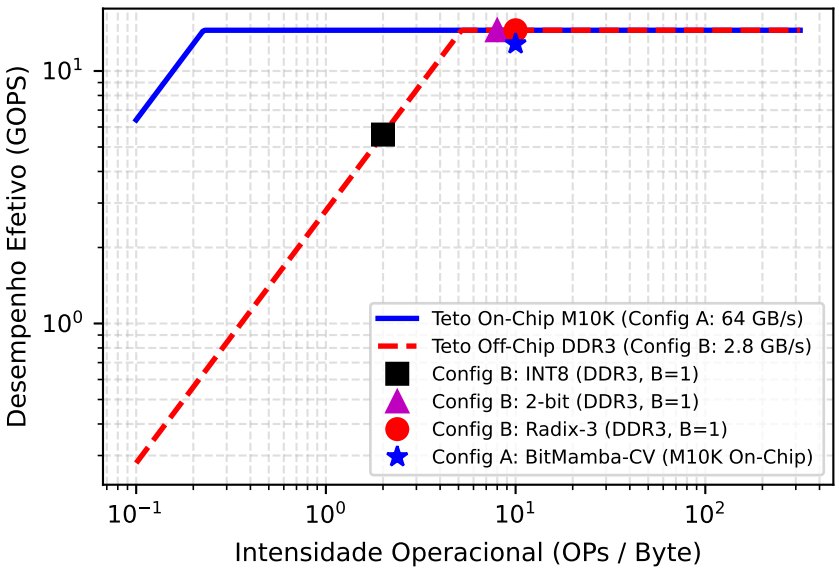
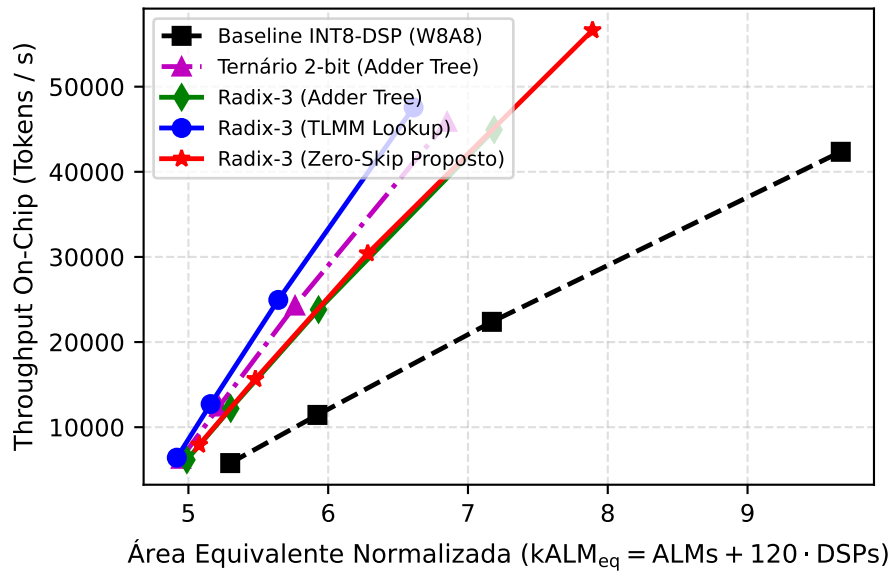

(a) Modelo Roofline Dual-Regime (Cyclone V)
Desempenho Efetivo (GOPS)
Intensidade Operacional (OPs / Byte)
10^1
10^0
10^{-1}
10^0
10^1
10^2
Teto On-Chip M10K (Config A: 64 GB/s)
Teto Off-Chip DDR3 (Config B: 2.8 GB/s)
Config B: INT8 (DDR3, B=1)
Config B: 2-bit (DDR3, B=1)
Config B: Radix-3 (DDR3, B=1)
Config A: BitMamba-CV (M10K On-Chip)


(b) Fronteira de Pareto Mesmo-Chip ($P \in \{8, 16, 32, 64\}$)
Throughput On-Chip (Tokens / s)
Área Equivalente Normalizada (kALM$_{eq}$ = ALMs + 120 · DSPs)
50000
40000
30000
20000
10000
5
6
7
8
9
Baseline INT8-DSP (W8A8)
Ternário 2-bit (Adder Tree)
Radix-3 (Adder Tree)
Radix-3 (TLMM Lookup)
Radix-3 (Zero-Skip Proposto)